

Home Task

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Subject: ADBMS

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Q.1

You are given three tables: Students, Friends and Packages. Students contains two columns: ID and Name. Friends contains two columns: ID and Friend_ID (ID of the ONLY best friend). Packages contains two columns: ID and Salary (offered salary in \$ thousands per month).

Column	Type
ID	Integer
Name	String

Students

Column	Type
ID	Integer
Friend_ID	Integer

Friends

Column	Type
ID	Integer
Salary	Float

Packages

Write a query to output the names of those students whose best friends got offered a higher salary than them. Names must be ordered by the salary amount offered to the best friends. It is guaranteed that no two students got same salary offer.

Sample Input

Congratulations

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Test case 0

Compiler Message

Success

Input (stdin)

1 INPUT

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Expected Output

1 Stuart
2 Priyanka
3 Paige
4 Jane
5 Julia

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Solution:

```
1  SELECT s.Name
2  FROM Students s
3  ∨ JOIN Friends f
4  |      ON s.ID = f.ID
5  ∨ JOIN Packages p1
6  |      ON s.ID = p1.ID
7  ∨ JOIN Packages p2
8  |      ON f.Friend_ID = p2.ID
9  WHERE p2.Salary > p1.Salary
10 ORDER BY p2.Salary;
```

Q2

You are given a table, Functions, containing two columns: X and Y.

Column	Type
X	Integer
Y	Integer

Two pairs (X_1, Y_1) and (X_2, Y_2) are said to be symmetric pairs if $X_1 = Y_2$ and $X_2 = Y_1$.

Write a query to output all such symmetric pairs in ascending order by the value of X. List the rows such that $X_1 \leq Y_1$.

Sample Input

X	Y
20	20
20	20
20	21
23	22
22	23
21	20

Congratulations

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Test case 0

Compiler Message

Success

Input (stdin)

1 INPUT

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Expected Output

1 2 24
2 4 22
3 5 21
4 6 20
5 8 18

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Solution

```
1  SELECT f1.X, f1.Y
2  FROM Functions f1
3  JOIN Functions f2
4  ON f1.X = f2.Y
5  AND f1.Y = f2.X
6  WHERE f1.X < f1.Y
7
8  UNION
9
10 SELECT X, Y
11 FROM Functions
12 WHERE X = Y
13 GROUP BY X, Y
14 HAVING COUNT(*) > 1
15
16 ORDER BY X;
```